

Aria Kidr

ariakidr@icloud.com | +44 7578793190 | Birmingham, UK | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

MSci Theoretical Physics, University of Birmingham Expected 2028 Current Weighted Average: 75% (First-Class Standard). Relevant Modules: Chaos (81%), Hamiltonian Mechanics (80%), Foundations in Data Science (79%).	Sep. 2024 – Jun. 2028
Sandwell Academy A-Level: Mathematics (A*), Physics (A) and Further Mathematics (A).	Sep. 2017 – Jun. 2024

RESEARCH EXPERIENCE

Independent Research Project <i>GPU-Accelerated Machine Learning for Derivatives Pricing</i> - Developing a quantitative-finance research project investigating neural surrogates for calibration of the Heston stochastic-volatility model. - Implementing and validating foundational pricing methods in Python, beginning with Black-Scholes valuation, implied volatility and Monte Carlo simulation. - Designing a reproducible PyTorch pipeline with planned evaluation of pricing accuracy, parameter recovery and computational performance.	Aug. 2026 - Present
University of Birmingham <i>Summer Research Intern</i> - Developed and optimised an autoencoder to denoise and reconstruct stellar power spectra, using controlled simulations and learning curves to refine the architecture. - Generated physically informed synthetic spectra and processed Kepler observations for approximately 5,000 red giants into reproducible training and validation datasets. - Analysed latent representations and investigated nonlinear recovery of the frequency of maximum oscillation power using reconstruction losses, residual diagnostics and signal-processing baselines. - Maintained documented code on GitHub and discussed the research at the 2026 National Astronomy Meeting.	Jun. 2026 – Aug. 2026 <i>Sun, Stars, and Exoplanets Group</i>
University of Birmingham <i>Summer Research Assistant</i> - Developed a configurable Python simulation of the 2D Ising model, implementing Hamiltonian, energy, magnetisation and local spin calculations. - Applied the Metropolis Monte Carlo across temperatures, initial conditions and Hamiltonian specifications. - Performed systematic parameter sweeps identifying 84 distinct low-energy configurations and analysed convergence through energy, magnetisation and lattice-evolution visualisations.	Jun. 2025 – Sep. 2025 <i>Theoretical Physics Group</i>

ADDITIONAL EXPERIENCE

MyTutor <i>Tutor</i> - Currently tutoring Mathematics and Physics at A-Level and GCSE, adapting explanations to individual needs. - Have had multiple students come back to me proud of their results coming back beyond target.	Oct. 2024 – Present <i>Remote</i>
DH Furniture Group Ltd <i>Furniture Assembler</i> Operated a CNC cutting machine and assembled furniture to technical specifications and quality standards.	Jul. 2022 – Sep. 2022 <i>Birmingham, Perry Barr</i>

TECHNICAL SKILLS AND ADDITIONAL INFORMATION

- Programming:** Python; NumPy, Pandas, SciPy, Matplotlib, TensorFlow/Keras, Jupyter, Git/GitHub.
- Quantitative Methods:** Machine learning, Monte Carlo simulation, Statistical modelling.
- Awards & Activities:** Birmingham Project Award for developing a proposal integrating Siemens technologies into sustainable campus education; completed five team-based Python, Bash and HPC challenges at the BLUEBear Challenge, including galaxy classification, ray tracing and supercomputer network-topology design.
- Languages:** English, Kurdish (Sorani) | **Interests:** Swimming, Badminton and Film Criticism.